Causal Diagnostics for Candidate Selection in Stochastic Transit Routing

(Authors' names are not included for peer review)

Abstract. A missed vehicle can reflect an omitted feasible connection, a poor choice among admitted connections, or a continuation forecast that assigns the wrong value to recovery. We distinguish these mechanisms using a committed boarding model with fixed vehicle outcomes, time-stamped local observations, and a common capped journey loss. Two exact multileg examples on Zürich timetable fragments isolate a loss from omitting a same-vehicle continuation and a loss from decision-making within a candidate set. A 35-date, 18-OD generated panel calibrated to route-identified forecasts gives mean losses of 27.05 minutes for a feasible nominal policy and 25.88 for a history-conditioned rollout. Their mean compute times are 3.59 and 16.24 seconds. A separate deadline stress panel slightly favors the nominal policy. Matched transition budgets on six further conditions show where smaller candidate menus can save time. A two-date measured railway replay and a held-out endpoint check delimit the realism of the generated vehicle model. The results support evaluating candidate admission and continuation scoring separately before committing to a candidate compression algorithm.

Key words: stochastic transit routing, partial information, recourse, evaluation

1. Decision problem and related work

A traveler has arrived at a stop, possibly on a vehicle that continues toward the destination. They may stay with that vehicle, change to a different one, or wait for information. The traveler's choice changes which events can be observed before the next commitment. A ranking that removes a boardable same-vehicle segment before scoring can therefore fail even when its conditional failure probabilities are accurate. Conversely, a candidate menu can contain both viable alternatives while their relative costs depend on the value of waiting or the action available after a missed departure.

Partial stop-level information and adaptive public transit policies have established precedents (Chen and Nie, 2015; Rambha et al., 2016). Geissmann and Gianinazzi (2019) give efficient adaptive policies for independently failing temporal connections, including a Zürich study. Durner (2024) studies realtime delay predictions, stochastic arrival labels, and several transfer strategies. These results motivate using a native strict strategy as a comparison and distinguishing its conditional arrival criterion from a capped journey loss. In a different setting, Hanasusanto et al. (2015) study a restricted inventory of adaptive plans, while Silver and Veness (2010) use simulation to search over action–observation histories. Here we use that style of search as a reference, with an explicit transition budget. Our contribution is the diagnostic comparison on a common execution

contract, including where candidate restriction, scoring and model fidelity change the interpretation of the results.

2. Execution contract and continuation values

Let t_0 be the departure clock and C the deadline in minutes from t_0. The loss is the elapsed time at destination if reached by $t_0 + C$, and C otherwise. Each generated world fixes one cancellation and one integer delay for each trip. The delay shifts the scheduled departure and arrival of each segment of that trip. Different trips are independent in the fitted city-panel model. The exact diagnostic fragments can instead couple outcomes by a persistent regime.

A public history h contains the stop, clock, last arrival, current trip, and observed vehicle events and attempt feedback. Each local decision publishes up to five unseen events that have occurred at the stop in the last ten minutes. A departure at the current minute remains boardable. A different vehicle requires the minimum transfer time from the previous arrival. Staying on the same trip incurs no transfer time. A boarding choice commits to at most 12 minutes of waiting after the traveler is ready, subject to the overall deadline. A one-minute wait defers the commitment. The scheduled departure lookahead is 25 minutes. Boarding, cancellation, a missed departure and an expired commitment generate different feedback. A missed attempt reveals that the vehicle departed too early. The exact departure becomes available only if it was independently reported. The execution clock uses physical travel and waiting times. We measure planner wall time separately.

For a first boarding action c, write B for boarding within the commitment window, $p = \Pr(B \mid h, c)$, A for the conditional expected capped loss given B, and F for the corresponding cost after a failed first attempt. The costs use the same causal continuation policy and include elapsed time from the common origin. Then

$$Q(h, c) = pA + (1 - p)F. \tag{1}$$

A known trip outcome conditions all of its segments. An unknown trip uses a discrete plug-in delay model conditioned on receipts, missing publications and attempt feedback. Route fits use online mean and cancellation updates and a historical delay spread. We integrate p directly on the departure support, and integrate A directly when the action reaches the destination. A shared bank of eight worlds estimates the other conditional costs. When a needed positive-mass branch is missing, one additional draw from that branch changes only the selected trip, retaining every base world. This exploits the conditional independence of the fitted trips.

The nominal policy uses conditional mean event times and a feasible timetable search with waiting. The rollout uses the same nominal policy as its tail, with the route parameters and candidate menu frozen for that query. Subsequent simulation decisions update their public trip information. The deployed policy can also refresh its route parameters and menu. The online shortlist prioritizes feasible same-trip continuations and admits up to five boardings and the wait action. An ablation replaces F by C for the current attempt. Every other transition and tail rule is shared in that comparison.

3. Candidate and information references

Let H specify an allowed action set at every future public history. With the same loss and information, a policy π restricted by H satisfies

$$J(\pi) - J^*_{\text{full}} = (J^*_H - J^*_{\text{full}}) + (J(\pi) - J^*_H). \tag{2}$$

The first term measures the cost of candidate restriction. The second is the cost of decisions within the rule. Exact finite-bank dynamic programming groups worlds by public outcome before selecting a continuation, and uses the same physical publication rule with either action menu. Knowing a regime changes the initial mixture while leaving several possible futures. Full hindsight discloses the world. With a fixed menu, increasing information can only weakly lower the optimum. This gives separate reference values for candidate restriction and information availability.

Two four-connection Zürich fragments illustrate the distinction. Removing the through service's same-trip continuation increases the same-information optimum from 3.625 to 5.75 minutes. A restricted heuristic adds 1.25 minutes. In a second fragment, deleting a transfer backup leaves the same-information optimum at 6 minutes, and the restricted heuristic costs 8. Knowing the current regime gives full and restricted costs of 5.6 and 5.8 minutes. Hindsight gives 5 minutes for both. The second example uses two possible future worlds within each known regime. Increasing its horizon from 12 to 30 minutes grows the full reference from 572 to 18,869 information nodes. This measures exact enumeration on a fragment with a specified world bank.

4. Forecast-calibrated generated comparisons

Exact trip identifiers link 189 trips and 2,000 connections to 33 route IDs. Sampling the archived realtime feeds and deduplicating event revisions produces 1,027,848 delay forecasts and 109,335 observed trip statuses over 35 dates. Of those statuses, 237 declare cancellation. Daily world parameters use the target date's delay forecasts and declared service statuses. The policies use

priors built from strictly earlier dates, with a documented fallback for missing route support. The timetable template deliberately holds topology fixed across conditions. Its constituent trips span different GTFS service calendars, which activate 74–176 of its connections per date in the archived static feed. The 35-date panel therefore describes generated trips on a common timetable template. Calendar-consistent measured evidence is reported separately below.

The main experiment evaluates the original 18 OD pairs for each date under one common generated world, using a 120-minute cap. The four strategies are an online marginal-label score, native Durner strict, the feasible nominal policy and the continuation rollout. All face the same exogenous vehicle outcomes and publication mechanism. Native strict keeps its own objective and is evaluated under the common capped loss. Table 1 summarizes 2,520 journeys. The rollout improves on nominal by 1.17 minutes on average, with 59 improvements, 15 regressions and 556 ties among the 630 OD conditions. Resampling the 35 generated date groups gives a descriptive rollout-minus-nominal interval of $[-2.49, -0.30]$ minutes. These generated groups are distinct from a sample of measured disruptions.

Table 1 Generated 35-date, 18-OD panel

Strategy	Mean loss (min)	Arrival (%)	Mean compute (s)
Marginal score	52.00	70.95	4.56
Durner strict	61.56	63.81	4.44
Feasible nominal	27.05	95.08	3.59
Continuation rollout	25.88	95.87	16.24

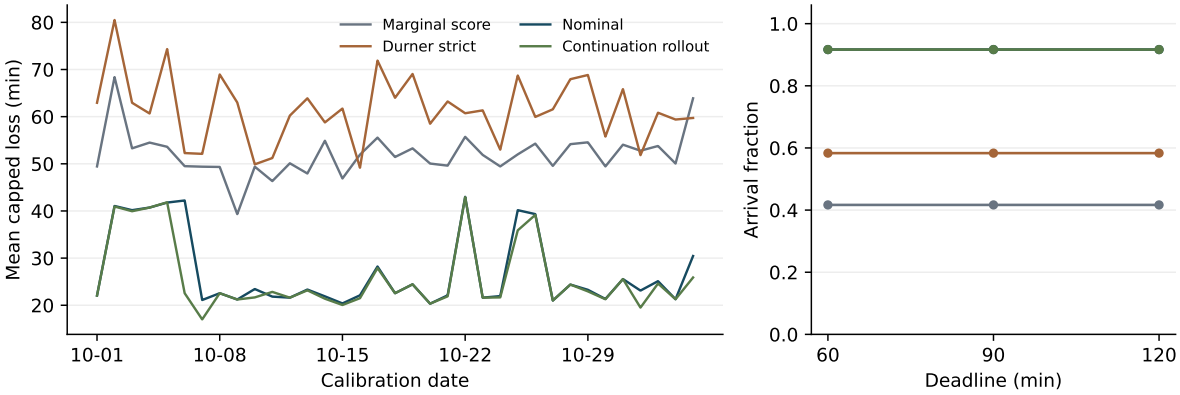
Timing includes initial preparation and all online decisions. Each date is one generated vehicle world shared across the 18 OD pairs.

On 12 separate stress conditions, the date-specific delay location is increased by eight minutes, the standard deviation has a four-minute floor, and the cancellation probability has a 20% floor. Across the 60-, 90- and 120-minute caps, nominal and rollout each arrive on 11 conditions and rollout costs 0.67 minutes more on average at each cap. In another eight generated conditions with new OD pairs and dates from the earlier forecast model, nominal costs 31.00 minutes and rollout costs 31.63. Figure 1 shows date variation and the deadline sensitivity. The comparison supports keeping nominal as a substantive, low-compute reference.

5. Matched computation and measured replay

On six route-identified conditions from three dates we compare unrestricted physical actions, a frozen initial menu, a smaller frozen menu, and a generic public-history UCT search. The smaller

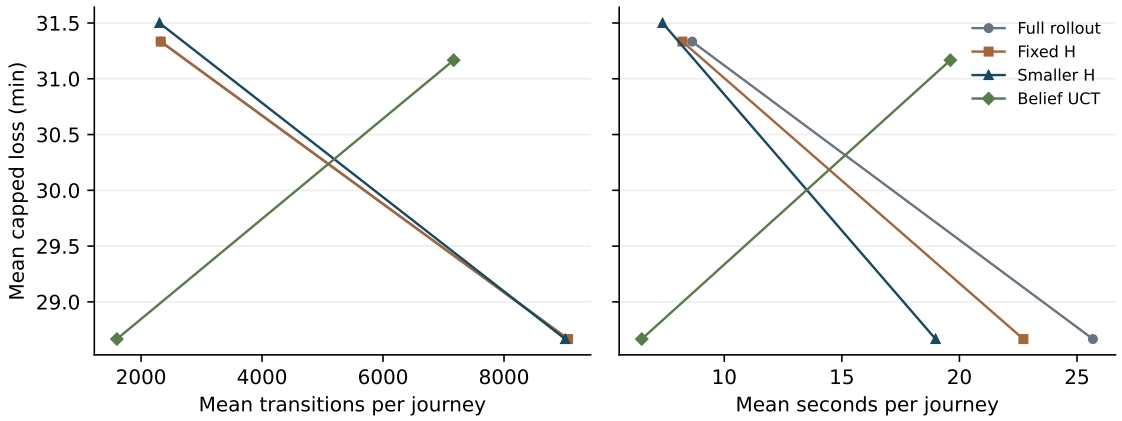
Figure 1 Date-level loss and arrival under deadline stress



Note. The left panel averages 18 OD pairs per generated world. The right panel uses 12 shared stress conditions per cap.

menu retains five initial labels per stop and closes the set under trip membership. It retains 1,149–1,508 of 2,000 connections. All planners respect a common upper limit of 128 or 512 committed-model transitions per decision. A rollout estimate uses only complete common-world rounds over the root actions and falls back to nominal when none completes. UCT allocates its transition budget over public histories. At budget 128 the full and smaller-menu mean losses are 31.33 and 31.50 minutes. At 512 they agree on all six journeys with mean 28.67. Mean end-to-end compute falls from 25.68 to 18.99 seconds at that larger budget. The matched limit counts transitions. Wall time also includes model setup, candidate preparation and search overhead. In these six conditions UCT has the lowest loss at 128 and is worse than full rollout at 512. This comparison measures a local empirical tradeoff for a bounded approximate search.

Figure 2 Quality and computation at two transition budgets



Note. Six paired generated conditions on the route-identified graph. The panels report actual simulated work and elapsed compute, respectively.

Official Swiss actual data (Open Data Platform Mobility Switzerland, 2023) provide unambiguous `REAL` timestamps for 54 SBB endpoints on October 12 and 25. On these endpoints, empirical 90% forecast intervals from earlier dates cover 19/30 and 8/24. Observed arrivals or departures beyond the upper interval endpoint explain most misses. All 27 measured complete legs have different departure and arrival delays, with a maximum difference of 239 seconds. That directly limits the common-shift assumption used in the generated comparison.

In a measured two-itinerary replay the traveler reaches Zürich HB on S11, then either stays aboard or transfers to S20 for Zürich Hardbrücke. Both connections are feasible with the specified transfer time on both dates. Staying on S11 reaches the destination 302 and 233 seconds earlier, respectively. We separately record the actual times and the forecast available before each decision. The two measured itineraries identify realized costs on those services. Citywide evaluation requires measured competing services and date-specific network coverage.

6. Implications

The exact examples show that loss from candidate admission and loss within a menu can vary independently. The generated city experiments show that a feasible nominal policy captures much of the improvement over the marginal score. Explicit rollout adds a smaller and computationally more costly increment on the main panel, and reverses direction on the stress and fresh screens. The budgeted experiments show a promising but limited cost reduction from a smaller menu. A city-scale bound on candidate loss and a cheap stopping certificate remain separate research questions. The measured endpoints call for an observation and delay model that allows within-trip variation before interpreting a generated-policy ranking as a deployment outcome.

Electronic Companion

Appendix A: First-event scoring and the policy gap

Equation (1) is the law of total expectation. For fixed continuation and nonempty branches, subtracting a fitted score gives the exact decomposition

$$\widehat{Q} - Q = (\widehat{p} - p)(A - F) + \widehat{p}(\widehat{A} - A) + (1 - \widehat{p})(\widehat{F} - F). \tag{3}$$

The conditionally sampled means estimate the fixed fitted tail, and the choice made by minimizing estimates can still have selection error. If a branch has zero mass its conditional cost is undefined and its contribution to the mixture is zero. No estimated probability alone identifies the effect of choosing a different future tail.

For the within-menu gap in (2), write the one-step optimal continuation under H as $Q_H^*(h,a) = \mathbb{E}[c + V_H^*(h') \mid h, a]$, with the same terminal cost for every policy. The sum of $Q_H^*(h,a) - V_H^*(h)$ along the executed policy telescopes in expectation to $J(\pi) - J_H^*$. Thus, if a greedy score has a uniform action error of at most $\delta(h)$ at each visited history, the per-step advantage is at most $2\delta(h)$. This is a conditional bound. The sampled fitted score supplies neither a uniform $\delta(h)$ nor a city-scale exact optimum.

Appendix B: Statistical and formalization scope

With a Normal–Gamma posterior using rate β_n, the marginal Student-t scale squared of the mean is $\beta_n/(\alpha_n\kappa_n)$, and its variance for $\alpha_n > 1$ is $\beta_n/[(\alpha_n - 1)\kappa_n]$. These differ from the predictive variance for a future delay. A fixed Lipschitz penalty gives a Wasserstein upper bound under the usual moment assumptions. An affine equality additionally requires a feasible translation in the support. This is standard structure (Mohajerin Esfahani and Kuhn, 2018). A separate Bayesian risk formulation requires an explicit parameter-risk functional (Lin et al., 2022). For a fixed ambiguity set, the worst-expectation map is nonexpansive in the supremum norm and discounting gives the discount factor as a contraction modulus. An undiscounted stochastic shortest path needs its own termination conditions. The finite reference here uses a finite horizon.

The archived Lean artifact checks abstract statements under their encoded hypotheses. In particular, its Lipschitz-penalized expectation is defined as that expression, and the posterior theorem uses a supplied scale or error condition. A source map in the artifact README records the Lean objects and their assumptions. Lean 4.28.0 checks all imported proof modules and the main module with `lake build +Main:olean`. The checked scope is the abstract statements. Simulator contracts, forecast calibration, runtime and the reported comparisons are supported by the executable tests and experiment receipts.

References

Chen P, Nie YM (2015) Optimal transit routing with partial online information. *Transportation Research Part B*. https://doi.org/10.1016/j.trb.2014.11.007.

Durner R (2024) *Stochastic Strategies for Public Transport Journeys Based on Realtime Delay Predictions*. Master's thesis, University of Stuttgart. https://doi.org/10.18419/opus-14802.

Geissmann B, Gianinazzi L (2019) Routing in stochastic public transit networks. *ATMOS*, article 4. https://doi.org/10.4230/OASIcs.ATMOS.2019.4.

Hanasusanto GA, Kuhn D, Wiesemann W (2015) K-adaptability in two-stage robust binary programming. *Operations Research*. https://doi.org/10.1287/opre.2015.1392.

Lin Y, Ren Y, Zhou E (2022) Bayesian risk Markov decision processes. *Advances in Neural Information Processing Systems*.

Mohajerin Esfahani P, Kuhn D (2018) Data-driven distributionally robust optimization using the Wasserstein metric. *Mathematical Programming* 171:115–166.

Rambha T, Boyles SD, Waller ST (2016) Adaptive transit routing in stochastic time-dependent networks. *Transportation Science*. https://doi.org/10.1287/trsc.2015.0613.

Silver D, Veness J (2010) Monte-Carlo planning in large POMDPs. *Advances in Neural Information Processing Systems*.

Open Data Platform Mobility Switzerland (2023) Actual data, October 2023 archive. https://archive.opentransportdata.swiss/istdaten/2023/ist-daten-2023-10.zip.